

Median, MAD, and Robust Sigma in `flag_folds()`

Why the method works for localized fold/lost-tissue artifacts, and why it breaks down when misregistration covers most of the image.

1. The problem the code is solving

For a pair of co-registered DNA rounds, each tile gets a disagreement score:

$$\text{disagree} = 1 - \text{NCC}(\text{tile}_a, \text{tile}_b)$$

NCC (normalized cross-correlation) is close to 1 when two tiles look alike (well-aligned, same tissue), so `disagree` is close to 0 for good tiles. A folded or lost-tissue tile looks different between rounds, so its `disagree` value is higher.

The question `flag_folds()` has to answer: **how high is "too high"**? You can't use one fixed number for every image, because different tissues/stains have different baseline noise levels. So the code tries to find "too high" *relative to what's typical for this specific image*.

2. Why median instead of mean

The obvious first idea: compute the mean and standard deviation of `disagree` across all tiles, then flag anything more than `k` standard deviations above the mean — the classic `mean + kσ` z-score approach.

The problem: **mean and standard deviation are themselves not robust to outliers**. If even a few tiles have a fold artifact with a very high disagreement score, those large values drag the mean upward and inflate the standard deviation. The threshold `mean + kσ` then rises to accommodate the very artifacts you're trying to detect — a self-defeating loop.

Median doesn't have this problem. The median is the middle value of a sorted list — it only cares about rank, not magnitude. Whether the top 10% of tiles have disagreement scores of 0.3 or 3.0, the median doesn't move, as long as they stay in the top 10%. This makes it a much more stable estimate of "what's typical" when a minority of values are contaminated by artifacts.

3. Why MAD instead of standard deviation

Once you have a robust center (median), you need a robust measure of spread — the equivalent of standard deviation. That's the **Median Absolute Deviation**:

$$\text{MAD} = \text{median}(|x_i - \text{median}(x)|)$$

In words: take every value's distance from the median, then take the median of *those* distances. Just like the median itself, this statistic is barely affected by a small number of extreme values — a handful of huge fold-artifact scores can't drag the MAD up the way they'd drag up a standard deviation (where you square the deviations, amplifying outliers even more).

The 1.4826 scaling factor

Raw MAD isn't on the same numeric scale as standard deviation, so comparing it directly to a z-score-style threshold wouldn't mean what you expect. For data that follows a normal (Gaussian) distribution, it turns out:

$$\sigma \approx 1.4826 \times \text{MAD}$$

This constant comes from the inverse of the normal distribution's cumulative function evaluated at 0.75 (the MAD of a standard normal distribution is exactly 0.6745, and $1/0.6745 = 1.4826$). Multiplying MAD by 1.4826 converts it into an estimate that behaves like a standard deviation *if* the "good" tiles were normally distributed — which lets you reuse the familiar "k sigma" intuition ($k=3 \approx$ "3 sigma event") even though you're using robust statistics under the hood.

4. Putting it together: the threshold

```
med    = median(disagree values)
mad     = median(|disagree - med|)
robust_sigma = 1.4826 * mad
threshold = med + k * robust_sigma

is_outlier = disagree > threshold
```

This says: "flag any tile whose disagreement is more than `k` robust-standard-deviations above the image's own typical (median) disagreement." It's self-calibrating — every image sets its own baseline, so you don't need one global threshold that works for every stain/tissue combination.

5. Why it breaks when overlap is very low

The entire method rests on one implicit assumption:

Assumption: the tiles affected by fold/lost-tissue are a *minority* of the image. The median and MAD are only "robust" up to a point — specifically, a statistic has a **breakdown point**, the fraction of contaminated data it can tolerate before it stops reflecting the "good" population. The median's breakdown point is 50%: if fewer than half your tiles are bad, the median still sits inside the "good" population. If *more* than half are bad, the median itself gets computed from the contaminated tiles.

When two images have very little overlap, the tiles showing high disagreement aren't a small minority anymore — they can be most of the image. What happens then:

Quantity	Normal case (localized fold)	Your case (little overlap)
<code>median(disagree)</code>	Reflects the many well-aligned tiles → stays low	Reflects mostly bad tiles → is itself high
<code>MAD</code>	Small — most tiles cluster near the median	Can be inflated, or can shrink if most tiles are <i>uniformly</i> bad (little spread among the bad tiles)
<code>threshold = med + k · σ</code>	Low — sits just above the good-tile noise floor	Already high, because <code>med</code> itself absorbed the artifact level
Result	Bad tiles stand out clearly above threshold	Bad tiles look "normal" relative to the (contaminated) baseline — they blend in

In short: **the baseline you're comparing against is computed from the same data you're trying to flag as abnormal**. When abnormal tiles dominate, "abnormal" and "typical" become statistically indistinguishable from each other — the method loses its own reference point.

6. Why lowering `k` doesn't really fix it

Lowering `k` makes the threshold stricter (closer to the median), so it eventually dips low enough to catch a heavily-contaminated image's outliers too. But `k` is a single global knob applied to *every* image, including images where the median/MAD were computed correctly from a healthy majority of good tiles. For those images, a smaller `k` just means a stricter cutoff on an already-correct baseline — it starts flagging normal biological or imaging-noise variation as if it were a fold artifact.

You end up choosing between two failure modes with the same parameter:

- **k too high:** misses the case where contamination is widespread (median already absorbed it)
- **k too low:** over-flags images where the baseline was fine to begin with

No single value of `k` can serve both regimes, because the two situations need fundamentally different information: one needs "is this tile unusual *within this image*," the other needs "is this whole image unusual to begin with."

7. The direction out

Core idea: stop asking the per-tile robust threshold to also detect gross, image-wide registration failure. Split it into two checks:

1. **Whole-image / coarse check first:** compute a single overall agreement score (e.g., global NCC, or the fraction of foreground tiles with reasonable NCC) between the two rounds. If this fails a fixed, absolute cutoff, classify the image as "registration failed" and stop — don't hand it to the per-tile fold detector, whose statistics assume a healthy majority.
2. **Per-tile $\text{median} + k \cdot \text{MAD}$ only runs on images that pass step 1** — i.e., images where you can trust that most tiles are genuinely well-aligned, so the localized-artifact assumption underlying median/MAD holds.

This way k only ever has to solve the problem it was designed for (finding a minority of local artifacts among a majority of good tiles), and the catastrophic low-overlap case gets caught by a separate, absolute rule that doesn't rely on the image's own (possibly contaminated) statistics.

Generated as a working explainer for `qc_tissue_fold_lost.py` — `flag_folds()` logic.